

Safety Guard MQTT DOCUMENT

Basic Instructions

a. Define

Variable name	Description
prefix	use for topic, default is "sg"
SN	device unique code
platform	Identifies that this topic is published by which client. Options: server / device
action	Customized according to actual application scenarios

b. MQTT Configuration

Username: **To be added**

Password: **To be added**

Protocol: **mqtt://** or **ws://**

c. Topic Rules

{prefix}/{SN}/{platform}/{action}

e.g : sg/3003004fba2394/device/alarm

d. Subdevice Code Map

Category	Code
Door Sensor	DC
PIR	PIR
Wireless smoke detector	WSD

Gas leakage sensor	GLS
Wireless outdoor alarm	WOS
Wireless indoor alarm	WIS
Water leak alarm	WLS
Remote control	CON
To bee added	

API

a. Device registration

I. Device Publish

Action	auth			
Description	This topic will be published after the device is started or reconnected to the Internet.			
Payload	<i>key</i>	<i>optional</i>	<i>type</i>	<i>Description</i>
	sign	Y	string	You can choose to use the specified encryption algorithm to encrypt sn to ensure the legitimacy of the request.

```

Topic:
sg/3003004fba2394/device/auth
Payload:
{
  "sign":"ec38cb30f28cf6ca1ebaf6c8790042da14a27dcb"
}

```

IIIIIIIIII. Server Publish

Action	auth			
Description	When the server receives this information, a data comparison is made and a token is sent to the device. All subsequent actions of the device will carry this token to ensure the validity of the data. if the device is disabled, the token can be set to empty. Time calibration services can also be provided here through the time_string field			
Payload	key	optional	type	Description
	token	Y	string	You can choose to use the specified encryption algorithm to encrypt sn to ensure the legitimacy of the request.
	time_string	N	string	YYYYmmddHHiiss e.g: 20240416165938

```

Topic:
sg/3003004fba2394/device/auth
Payload:
{
  "token":"f28cf6ca1ebaf6c879004",
  "time_string":"20240416165938"
}

```

b. Device heartbeat

I. Device Publish

Action	heartbeat			
Description	The device will publish this topic approximately every minute to inform the server whether it is currently available.			
Payload	key	optional	type	Description
	token	N	string	
	ver	N	number	vercode
	list	N	array	subdevice list
	list->id	N	string	subdevice id Within the same host, this id is unique
	list->state	N	string(8)	subdevice's state state[0]=1 means offline state[5]=1 means the device triggers tamper state[6]=1 means the device has low power state[7]=1 means the door sensor is opened If everything is OK, 00000000 will be returned

```

Topic:
sg/3003004fba2394/device/heartbeat
Payload:
{
  "token": "f28cf6ca1ebaf6c879004",
  "ver":1001,
  "list": [
    {
      "id":"00411400",
      "state":"00000000"
    }
  ]
}

```

IIIIIIII. Server Publish

Action	heartbeat			
Description	The server should inform the device whether it can continue to work.			
Payload	<i>key</i>	<i>optional</i>	<i>type</i>	<i>Description</i>
	valid	N	number	0 - invalid device 1 - valid device

	list->code	N	string	Subdevice Code Map
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```

Topic:
sg/3003004fba2394/device/syncSubdevice
Payload:
{
  "token":"f28cf6ca1ebaf6c879004",
  "list": [
    {
      "id": "009fcbe0",
      "name": "DC demo",
      "area_name":"bedroom",
      "code": "DC"
    }
  ]
}

```

IIIIIIIIII. Server Publish

Action	syncSubdevice			
Description	General response			
Payload	<i>key</i>	<i>optional</i>	<i>type</i>	<i>Description</i>

```
Topic:
sg/3003004fba2394/server/syncSubdevice
Payload:
{ }
```

d. Delete subdevice

I. Device Publish

Action	delSubdevice			
Description	When the device deletes the subdevice, this topic will be published.			
Payload	<i>key</i>	<i>optional</i> /	<i>type</i>	<i>Description</i>
	token	N	string	
	ids	N	array(string)	subdevice id collection If the user chooses to delete all, this array will only have one member, and the value is ffffffff (length is 8)

```
Topic:
sg/3003004fba2394/device/syncSubdevice
Payload:
{
  "token":"f28cf6ca1ebaf6c879004",
  "ids": ["ffffffff"]
}
```

IIIIIIII. Server Publish

Action	delSubdevice			
Description	General response			
Payload	<i>key</i>	<i>optional</i>	<i>type</i>	<i>Description</i>

```

Topic:
sg/3003004fba2394/device/alarm
Payload:
{
    "token":"f28cf6ca1ebaf6c879004",
    "id": "00000040",
    "message":"Accessory disassembly!:door sensor is
triggered,Location:door",
    "name":"door sensor",
    "type":0
}

```

IIIIIIII. Server Publish

Action	alarm			
Description	no need response If you need the server to notify the alarm host to alarm, you can publish this topic			
Payload	<i>key</i>	<i>optional</i>	<i>type</i>	<i>Description</i>

```
Topic:
sg/3003004fba2394/server/alarm
Payload:
{ }
```

f. ChangeArmMode

I. Device Publish

Action	changeArmMode			
Description	If the alarm host or sub-device generates an alarm, this topic will be published.			
Payload	<i>key</i>	<i>optional</i> /	<i>type</i>	<i>Description</i>
	token	N	string	
	mode	N	number	0 - disarm 1 - away arm 2 - stay arm
	way	N	number	If 6, it means sync

```
Topic:
sg/3003004fba2394/server/changeArmMode
Payload:
{
    "mode": 1
}
```

g. OTA

I. Device Publish

Action	ota			
Description	When the user touches the Check for updates button on the alarm, this topic will be published			
Payload	<i>key</i>	<i>optional</i> /	<i>type</i>	<i>Description</i>
	token	N	string	
	ver	N	number	current version
	platform	N	number	Used to isolate different firmware

```

Topic:
sg/3003004fba2394/device/ota
Payload:
{
    "token":"f28cf6ca1ebaf6c879004",
    "ver": 1001
}

```

IIIIIIII. Server Publish

Action	ota			
Description	response from device ota or The server wants to provide upgrades to the specified device			
Payload	<i>key</i>	<i>optional</i>	<i>type</i>	<i>Description</i>
	url	N	string	http address(not https)
	length	N	number	file total bits

Topic:

sg/3003004fba2394/server/otaProgress

Payload:

```
{  
  "progress": 5  
}
```